

Julio Calvo

Frontend Engineer · Toronto, ON

juliocalvorios.com · +1 437-254-9832 · juliocalvorios@gmail.com

github.com/juliocalvorios · linkedin.com/in/juliocalvorios

TECHNOLOGIES

React · TypeScript · Next.js · Tailwind CSS · Supabase · Jest · Playwright · Three.js · Zod · React Hook Form

MAIN PROJECT

veraOS

veraos.ai

- Built 10 interactive widget types with drag-and-drop using $O(n)$ collision detection via spatial indexing
- Implemented AES-256-GCM encryption with server-side keys via Supabase Vault, zero client exposure
- Designed chat library system with nested folders, drag-and-drop organization, and optimistic updates with rollback
- Developed flashcard system with FSRS-4.5 spaced repetition algorithm and 8 card types
- Architected lazy-loading system with code splitting (5.3MB split into 50+ chunks that load on-demand)
- Offloaded embeddings processing to server-side (25MB → 0MB client download)
- Integrated 6 LLM providers with streaming semantic highlighting and automatic fallbacks

ADDITIONAL PROJECTS

Ontario Flag Time Machine — 73 unit tests + 5 E2E, TypeScript strict, branded types ontario-flag-project.vercel.app

The Impossible Form — React Hook Form + Zod, pixel-perfect Windows 95 UI the-impossible-form.vercel.app

Through the Glass — Three.js particles, GLSL shaders, real-time weather API through-the-glass-project.vercel.app

OPEN SOURCE

react-ai-highlight-parser

npmjs.com/package/react-ai-highlight-parser

Streaming text parser with $O(n)$ tokenization, 60 semantic highlight combinations

EDUCATION

Humber Polytechnic, Toronto — Diploma in Multimedia Design & Development 2024–2025

Montessori College, Spain — Commerce & Marketing Management 2019–2023